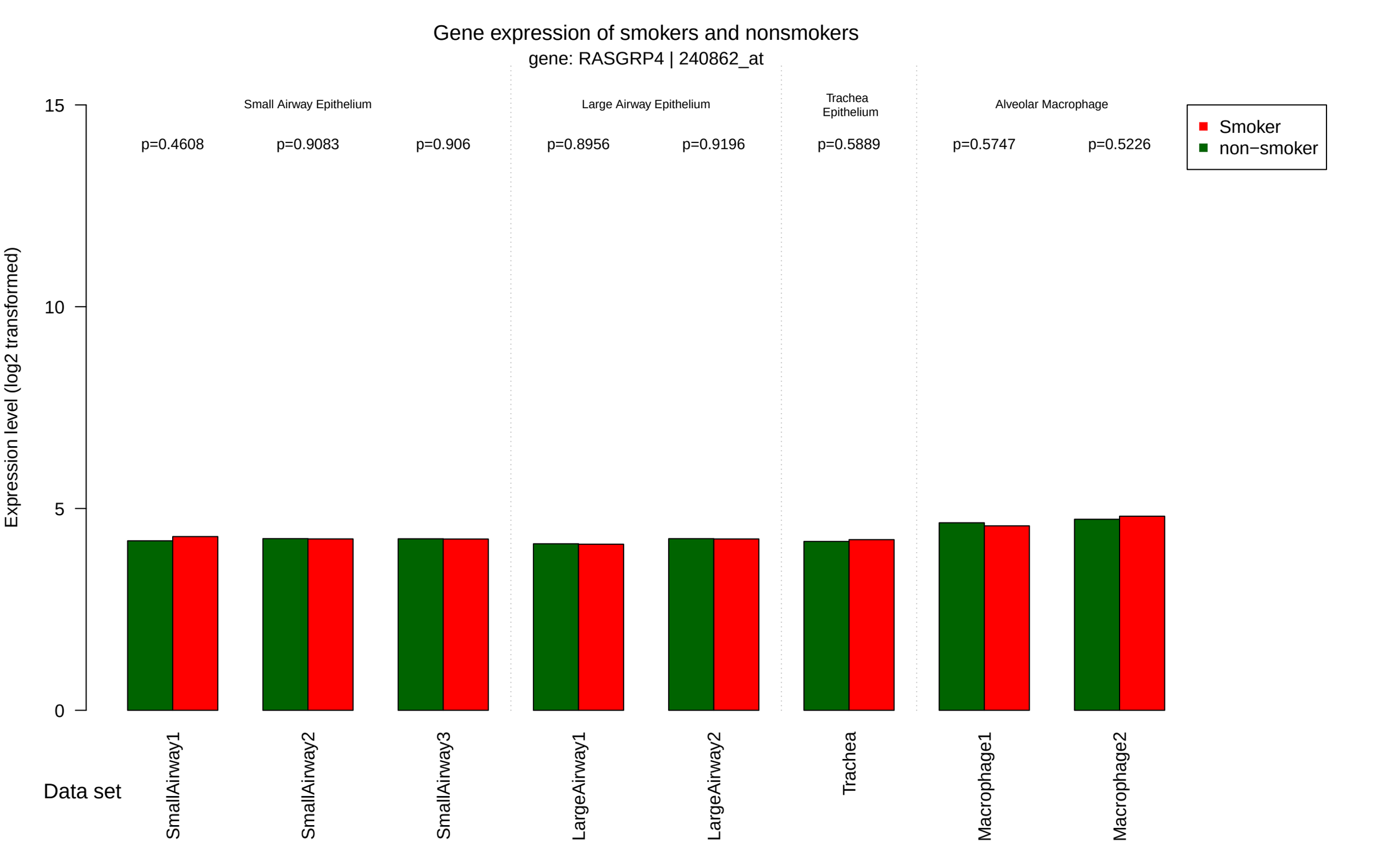
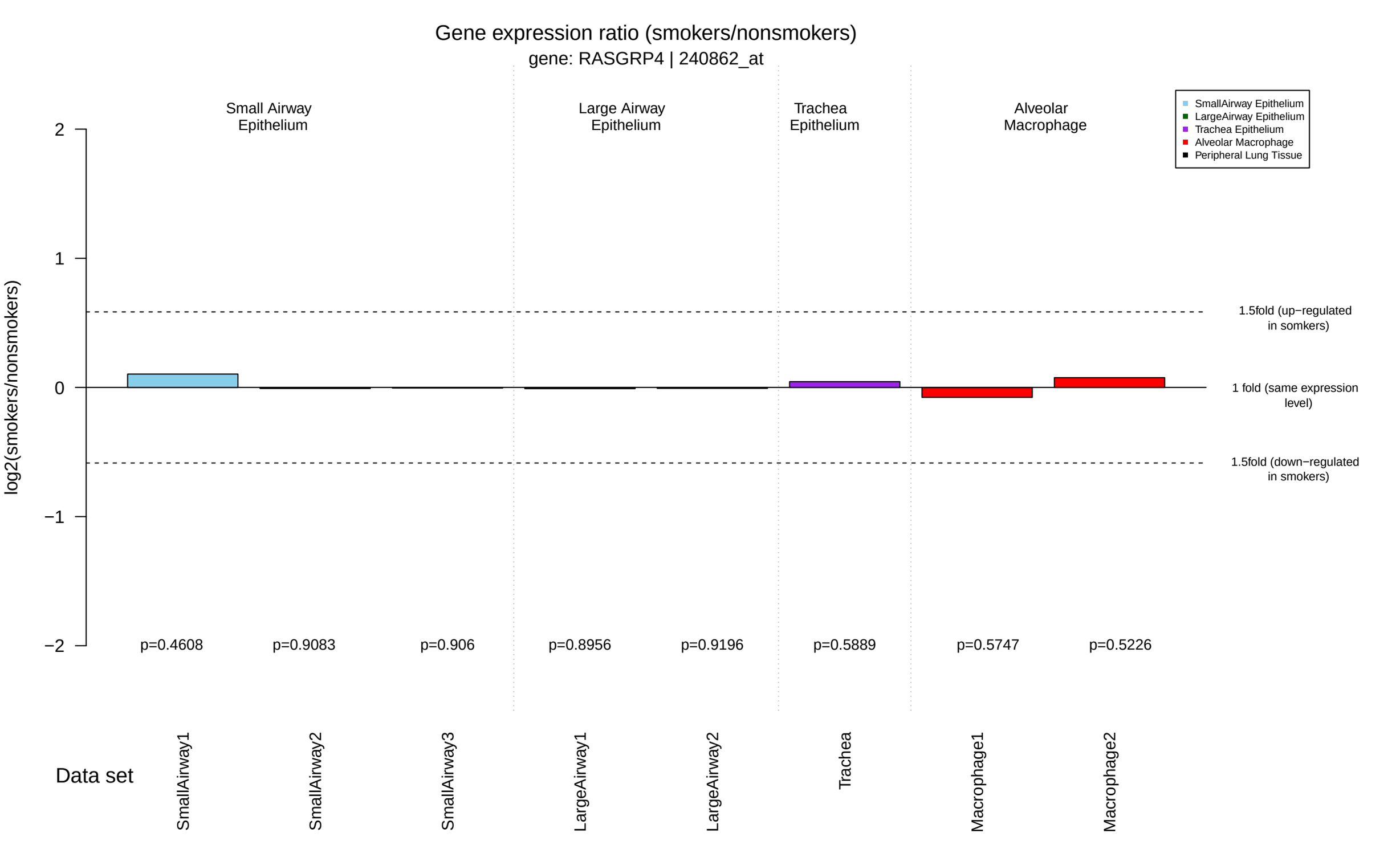
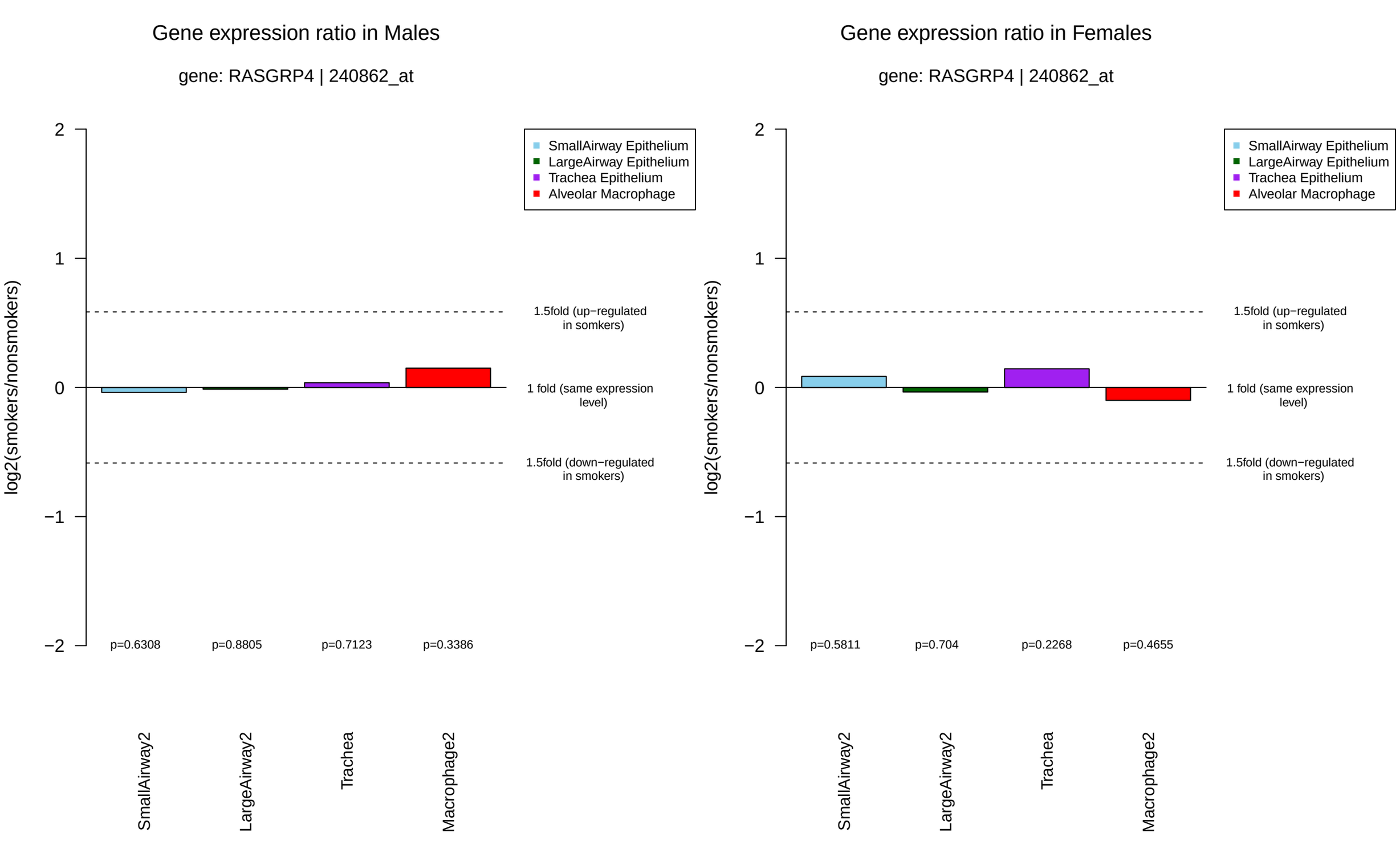
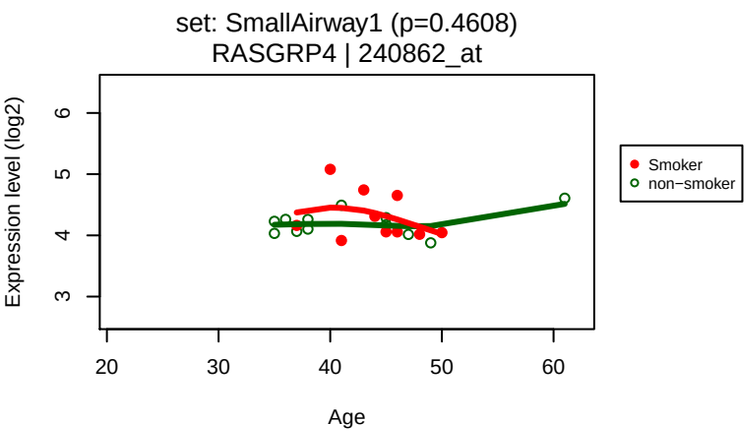
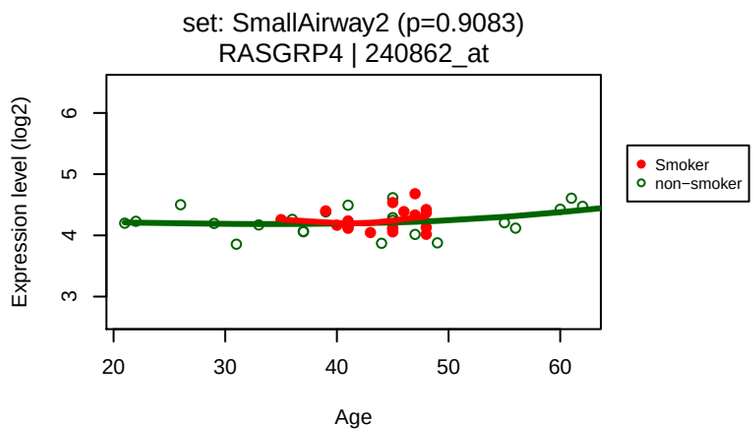
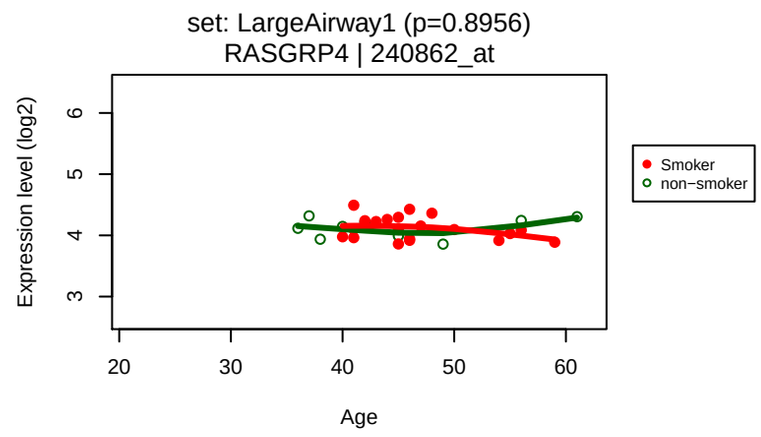

Gene expression of smokers and nonsmokers
gene: RASGRP4 | 240862_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4608
p=0.9083
p=0.906
p=0.8956
p=0.9196
p=0.5889
p=0.5747
p=0.5226
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

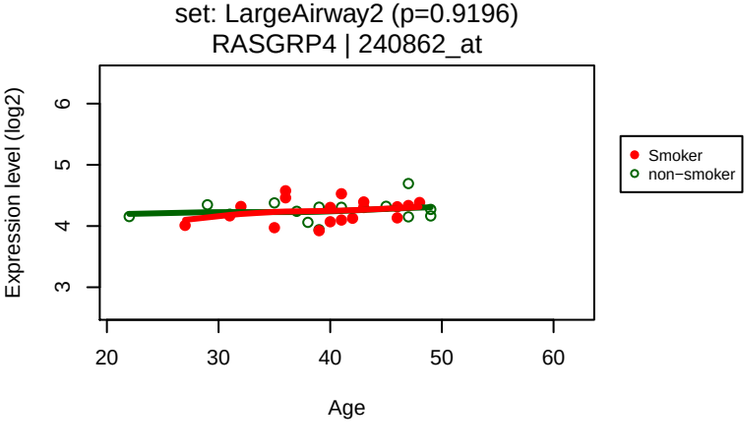

Gene expression ratio (smokers/nonsmokers)
gene: RASGRP4 | 240862_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4608
p=0.9083
p=0.906
p=0.8956
p=0.9196
p=0.5889
p=0.5747
p=0.5226
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

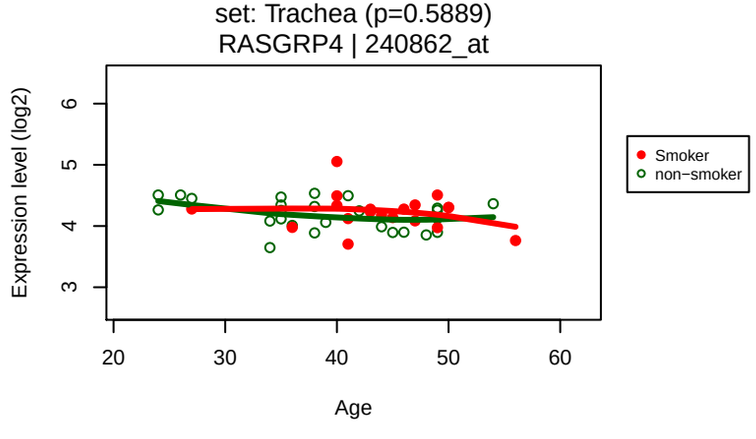

Gene expression ratio in Males
gene: RASGRP4 | 240862_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6308
p=0.8805
p=0.7123
p=0.3386
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASGRP4 | 240862_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5811
p=0.704
p=0.2268
p=0.4655
SmallAirway2
LargeAirway2
Trachea
Macrophage2

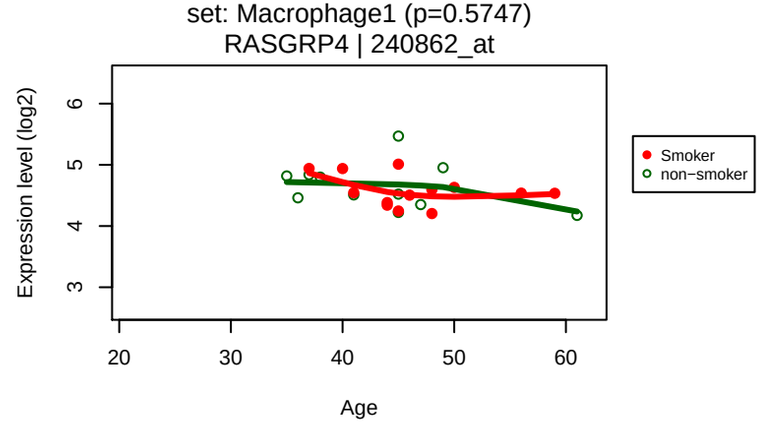

set: SmallAirway1 (p=0.4608)
RASGRP4 | 240862_at
Expression level (log2)
Age
Smoker
non−smoker

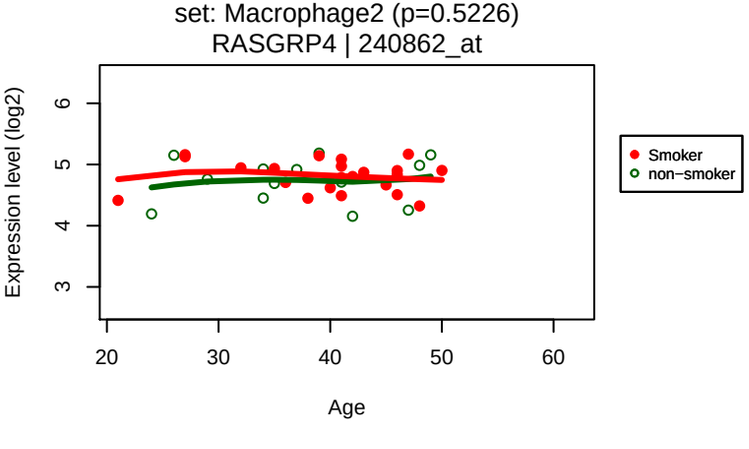

set: SmallAirway2 (p=0.9083)
RASGRP4 | 240862_at
Expression level (log2)
Age
Smoker
non−smoker

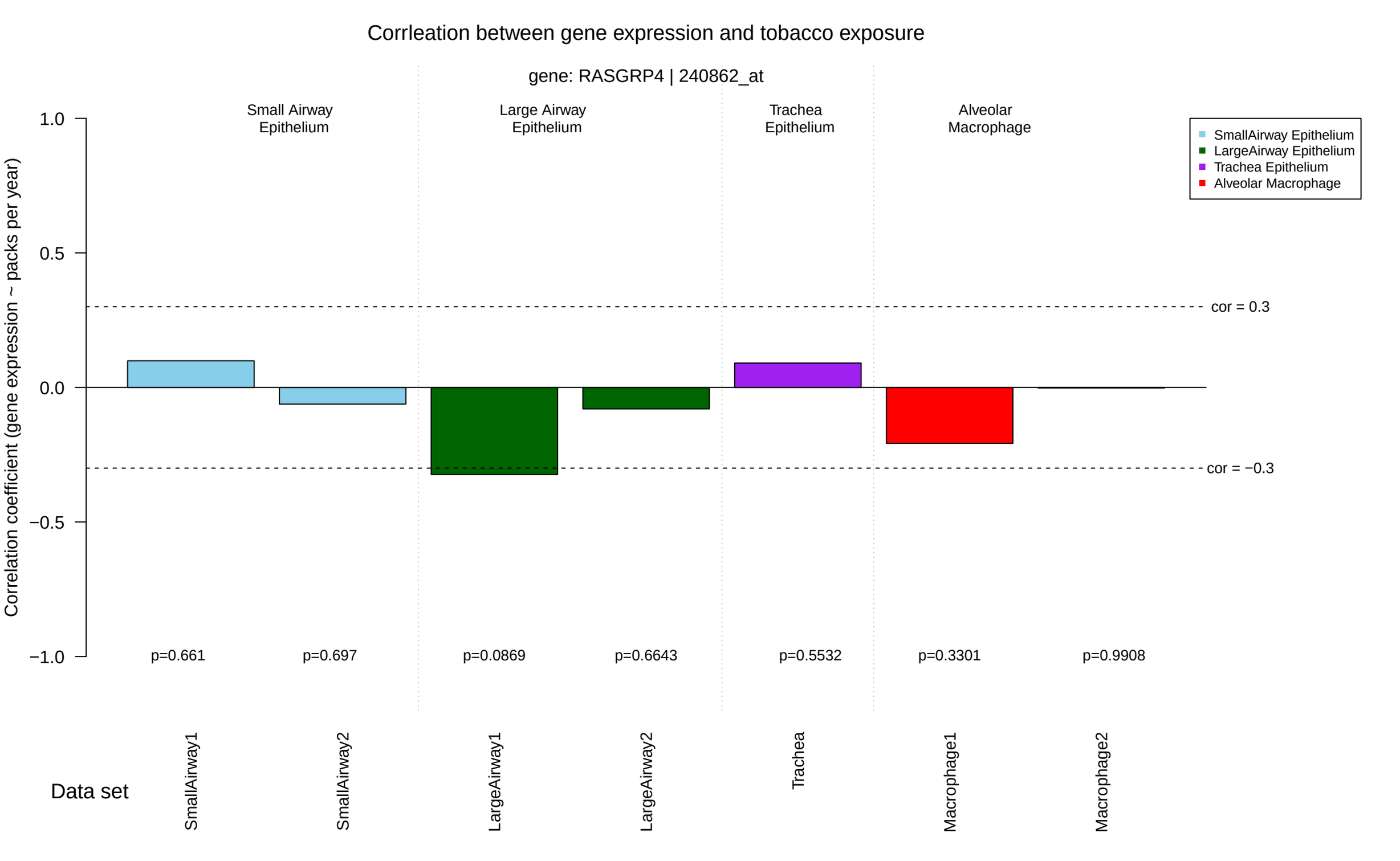

set: LargeAirway1 (p=0.8956)
RASGRP4 | 240862_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9196)
RASGRP4 | 240862_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5889)
RASGRP4 | 240862_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5747)
RASGRP4 | 240862_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5226)
RASGRP4 | 240862_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASGRP4 | 240862_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.661
p=0.697
p=0.0869
p=0.6643
p=0.5532
p=0.3301
p=0.9908
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2